



Supermassive black holes in the early universe

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ABSTRACT

The emergence of the first black holes during the first billion years of cosmic history marks a key event in cosmology. Their formation is part of the overall process of ending the cosmic dark ages, when the first stars appeared in low-mass dark matter haloes about a few 100 million years after the Big Bang. The first stars, galaxies, and black holes transformed the Universe from its simple initial state into one of ever increasing complexity. We review recent progress on our emerging theoretical picture of how the first black holes appeared on the cosmic scene, and how they impacted the subsequent history of the Universe. Our focus is on supermassive black holes, in particular assessing possible pathways to the formation of the billion-solar-mass black holes inferred to power the luminous quasars at high redshifts. We conclude with a discussion of upcoming empirical probes, such as the *James Webb Space Telescope* (JWST), and the Laser Interferometer Space Antenna (LISA), further ahead in time.

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1. Introduction

One of the key questions in modern cosmology is to understand when black holes first appeared on the cosmic scene, and when they began to impact the evolution of the Universe. Black holes are extreme objects, left behind when massive stars die, with a gravitational force that is so strong that nothing can escape, not even light [1]. The first black holes in particular played an important part in ending the cosmic dark ages, the period in the history of the Universe before any stars or galaxies had formed [2,3]. The dark matter, the dominant form of mass in the Universe, was initially distributed very uniformly, but was endowed with small fluctuations in its density, which in turn originated in quantum processes in the moments briefly after the Big Bang. These primordial perturbations were slowly amplified by gravity, to the point where some regions of dark matter could break away from the overall expansion of the Universe. Held together by their own gravity, those regions gave rise to roughly spherical dark matter ‘haloes’, which also contained trace amounts of normal matter, the hydrogen and helium gas made in the Big Bang.

Some haloes were able to host the first stars and galaxies, formed out of primordial gas. Within the standard model of how structures formed in the Universe, assuming the presence of dark matter and dark energy, this epoch of first light is predicted to have taken place a few 100 Myr after the Big Bang. In cosmology, it is convenient

to express ages in terms of the ‘redshift’, denoted by the symbol z , which measures how much the wavelength of light has been stretched by the expanding Universe for present-day observers. Thus, a larger redshift implies earlier times, closer to the Big Bang, such that the time when the first stars formed corresponds to $z \sim 20\text{--}30$. This period marks the fundamental transition from the simple initial conditions in the very early Universe to the ever increasing complexity that we see around us today (Table 1).

From a theoretical viewpoint, the formation of the first stars marks the moment when differences in density became increasingly pronounced. Previously, the growth of the small density perturbations was governed by simple linear equations, describing a Gaussian random field, with parameters calibrated to exquisite precision by measurements of the cosmic microwave background (CMB). Here, key probes of the CMB were the satellite-borne *Wilkinson Microwave Anisotropy Probe* (WMAP) and *Planck* missions. Afterwards, studying the emerging structures requires sophisticated numerical simulations, tracing both the dark matter and baryonic (gaseous) components. The first stars¹ also initiated the transition from primordial gas to one enriched with heavy chemical elements (‘metals’ in the parlance of astronomy), thus providing more efficient cooling channels that were not accessible in the early Universe [4,5].

Table 1. Summary of acronyms used throughout the review.

Acronym	Explanation
21 cm	Hyperfine-structure transition of neutral hydrogen
AGN	Active galactic nuclei (galaxy centres with accreting SMBHs)
CMB	Cosmic microwave background (created $\sim 380,000$ yr after the Big Bang)
DCBH	Direct-collapse black hole (massive seed without fragmentation)
H ₂	Molecular hydrogen (primary coolant at low temperatures)
HMXB	High-mass X-ray binary (black hole accreting from companion star)
IGM	Intergalactic medium (low density gas between galaxies)
<i>JWST</i>	<i>James Webb Space Telescope</i> (infrared wavelengths)
LIGO	Laser Interferometer Gravitational-wave Observatory
LISA	Laser Interferometer Space Antenna (gravitational-waves)
LW	Non-ionising (soft-UV) Lyman-Werner photons
Ly α	Lyman-alpha line of atomic hydrogen ($2p \rightarrow 1s$ transition)
Pop III	Primordial first stars (formed a few 100 Myr after the Big Bang)
SKA	Square Kilometre Array (radio interferometer)
SMBH	Supermassive black hole ($M_{\bullet} \gtrsim 10^6 M_{\odot}$)

The focus of this review is on supermassive black holes (SMBHs) in the early Universe, with masses² of $M_{\bullet} \gtrsim 10^6 M_{\odot}$, but it is likely that the first black holes were less massive, emerging from the collapse of Pop III stars [6]. Such stellar remnants would have masses of a few $\sim 10 M_{\odot}$. A fraction of these stellar-mass black holes might give rise to a population of tightly-bound double stars, called high-mass X-ray binaries (HMXBs), where the black hole is in orbit around a normal star. These sources release copious amounts of high-energy (X-ray) radiation when gas from the companion star is falling onto the black hole in the binary system, experiencing ferocious frictional heating in the process. This heat energy could in turn significantly influence the early intergalactic medium (IGM), through widespread X-ray ionisation and heating [7,8].

Another intriguing class of black holes may have formed in the ultra-early Universe, in the chaotic conditions that prevailed briefly after the Big Bang. Typically, such ‘primordial black holes’ are predicted to have masses that are much smaller than stars, extending all the way down to the Planck mass, $m_P \sim 10^{-8}$ kg, the quantum-gravitational unification scale³. Recently, a variant of primordial black holes has been proposed as the origin for the elusive dark matter [9]. To account for the dark matter, a sub-class of primordial black holes could have formed with typical masses of a few tens of solar masses, in line with the masses of the binary black hole mergers observed by the Laser Interferometer Gravitational-wave Observatory (LIGO). Since these objects would have formed before the epoch of Big Bang nucleosynthesis, when the primordial hydrogen and helium was produced, they would not have shown up in the baryon budget encapsulated in the abundance ratios of the primordial chemical species⁴.

We know that the Universe was able to form SMBHs with $M_{\bullet} \gtrsim 10^9 M_{\odot}$ in the centres of giant galaxies. Spectacular confirmation has recently come from the direct imaging of the immediate vicinity of such an object in the Messier 87 (M87) galaxy with the Event Horizon Telescope. This telescope consists of radio antennae that are distributed all over the Earth, thus providing the incredible power to resolve the central black hole in M87. Furthermore, observations of luminous, extremely distant quasars imply that SMBHs were already in place at very early times, corresponding to redshifts $z \gtrsim 7$ [10–13]. Quasars are vastly scaled-up cousins to the HMXB sources mentioned above, where again huge amounts of heat energy are released when surrounding gas falls onto the central black hole.

The question then arises how lower-mass black holes, acting as seeds, could have grown into these massive structures in the limited time available. Traditionally, stellar-mass seeds were assumed, possibly the remnants of massive Pop III stars. Those seeds with up to $M_{\bullet} \sim 100 M_{\odot}$ could then grow through mass assembly, or ‘accretion’, to reach the SMBH target mass by $z \sim 7$, as required by the observations [14]. There is a limit to the rate of how fast this assembly could have occurred, first worked out by the British astrophysicist Arthur Eddington. The idea here is that the higher the mass assembly rate, the larger the output of radiation. At some point the outward pressure exerted by the radiation is able to overwhelm the inward force of gravity. This balancing of forces defines the Eddington (accretion) rate, the maximum growth rate of black holes.

The stellar-seed scenario has been challenged, however, when taking into account the effects of energy input from stars on the efficiency of the accretion flows [15,16]. Specifically, the ultra-violet (UV) radiation from the massive black hole progenitor stars will heat the surrounding gas, thus driving strong outflows. As a consequence, the newly formed black hole finds itself, at least initially, in a low-density environment, taking time to replenish the gas supply in its vicinity. Mass growth rates, therefore, are greatly suppressed below the Eddington rate for about the age of the Universe at that time (~ 100 Myr). Under such conditions, stellar seeds require growth rates that are somehow able to exceed the Eddington rate to reach the SMBH target mass [8].

In response to this ‘timing crisis’, the alternative scenario of direct-collapse black holes (DCBHs) has been suggested [17,18]. The idea here is to trigger the collapse of a primordial gas cloud into a so-called ‘atomic cooling halo’, whose gravitational potential well is deep enough to heat the infalling gas to temperatures of $T \gtrsim 10^4$ K. Collisions between gas particles are thus sufficiently energetic to excite hydrogen atoms from the ground state.

The subsequent recombination radiation is then able to radiate away the heat energy, generated during the compression of the gas. The strongest cooling channel is often provided by the atomic hydrogen Lyman-alpha ($\text{Ly}\alpha$) line, when the atom returns from its first excited state into the ground state. To activate this ‘atomic cooling’ mechanism, the dark matter halo has to have a mass of $M_{\text{halo}} \sim 10^8 M_{\odot}$ at redshifts $z \gtrsim 10$.

Normally, such collapse would be accompanied by vigorous fragmentation, and thus star formation. The star-forming activity in turn would trigger strong energy input, which would act to dispel the gas supply from the host halo. To accomplish the assembly of a significant fraction of the gas into a supermassive object at the halo centre, the DCBH model assumes that any fragmentation is suppressed by preventing the gas from cooling down. Molecular hydrogen (H_2), different from the atomic hydrogen in $\text{Ly}\alpha$ emission, would enable such low-temperature cooling, because of the low-energy rotational and vibrational quantum states that can be excited. Destroying the H_2 may be possible if the primordial cloud is suffused in a very strong flux of soft-UV photons with energies just below the threshold to ionise hydrogen, but capable of dissociating molecular hydrogen⁵. Recent cosmological simulations have confirmed that such ‘collapse without fragmentation’ can indeed lead to the build-up of massive central structures, as we will discuss further below.

A population of black holes in the early Universe would result in the build-up of a cosmic X-ray background, a near-uniform, pervasive radiation field in the X-ray band. Of particular importance might be the HMXB sources mentioned above, possibly resulting from the first stars [19]. Their emission could contribute to the widespread ionisation and heating of the primordial gas, given the long mean free paths of the high-energy X-ray photons [20]. Prior to what cosmologists call the ‘epoch of reionization’ about a billion years after the Big Bang, the intergalactic gas was still substantially neutral, but any degree of (partial) ionisation from the X-ray background would impact the primordial chemistry, in particular the pathways towards forming H_2 , the main coolant in low-temperature metal-free gas [21].

Another important thermal effect of an early X-ray background may be the way it affects the coupling of the hyperfine-structure levels of neutral hydrogen to the kinetic energy of the cosmic gas. Those levels in turn determine the strength of the 21 cm spin-flip transition, which could be observed with dedicated radio telescopes [22,23]. Understanding the detailed physics behind this coupling is a key challenge in the emerging field of 21 cm cosmology, with pathfinder experiments such as the Experiment to Detect the Global Epoch of

Reionization Signature (EDGES) already underway [24], and with the large-scale Square Kilometre Array (SKA) on the horizon.

We have a number of tantalising windows from observations into the properties and impact of black holes at high redshifts. Next to the constraints from the most distant, hyper-luminous quasars on the growth of the first SMBHs, there are hints of a significant correlation between the unresolved cosmic X-ray and (near-) infrared backgrounds, possibly indicating a physically related origin, such as a population of massive Pop III stars [25]. Another promising probe of the first stellar-mass black holes are the gravitational waves emitted by binary black hole mergers. Estimates indicate that about 1% of the black hole merger events detected by LIGO could originate in Pop III progenitors [26]. In the near future, the direct hunt for the first SMBHs will be carried out by the *James Webb Space Telescope (JWST)*, with a particular focus on searching for DCBH sources. Once we have identified such promising SMBH candidates with the *JWST*, deep spectroscopic follow-up with the upcoming suite of extremely large telescopes on the ground will provide key diagnostics for the physical nature of these *titans of the early Universe* (see [27]).

The outline for this review is as follows. We begin with a discussion of the underlying black hole formation physics, within the context of standard cosmology, covering the entire range of possible masses. We next address the multiple ways in which black holes can impact the early IGM, acting both as drivers of cosmic evolution and providing empirical probes of the sources at work here. After assessing the potential for such probes in testing our emerging theoretical picture, we will conclude with our outlook for the future. Many of these themes consistently appear in reviews of this subject, albeit with particular emphases and continuous refinements. For further exploration, we refer the reader to this rapidly expanding literature [27–31].

2. Formation

There are several pathways to forming astrophysical black holes, including both canonical and exotic scenarios. We summarise the most relevant ones for the high-redshift Universe with the diagram in Figure 1 with further details in the following subsections. This roadmap provides the context for understanding the challenges encountered in order to grow efficiently and make connections with observational studies, especially in the context of the aforementioned quasar-seed timing problem. Early simulations of collapsing primordial gas clouds showed that most of the gas fragments into dense stellar clumps which eventually virialize into a spheroidal galactic bulge [33].

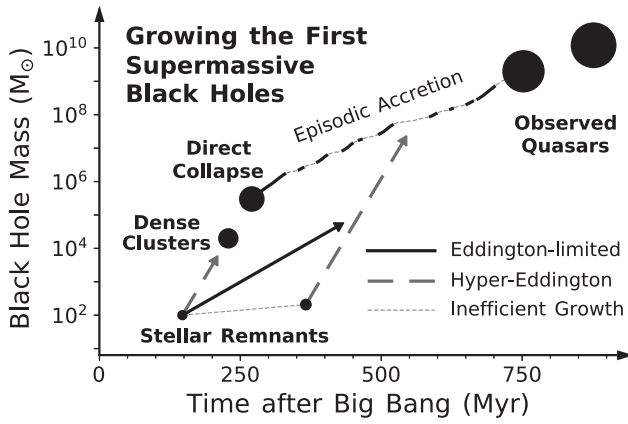


Figure 1. Illustrating the timing crisis for supermassive black hole growth in the early Universe. The observed quasars with masses of $M_{\bullet} \approx 10^9 - 10^{10} M_{\odot}$ at redshifts of $z \approx 6-7$ most likely originated from the following scenarios: the direct-collapse of a primordial gas cloud for which cooling does not initiate fragmentation, dense star clusters in which runaway collisions trigger black hole formation, and stellar-remnant black holes experiencing hyper-Eddington mass growth. The subsequent growth is dominated by episodic galaxy mergers and infall from streams of cold gas along dark matter filaments (adapted from [32]). © Aaron Smith.

Even without fragmentation, conservation of angular momentum during collapse generates a centrifugal barrier inhibiting rapid mass accretion [34,35]. However, the presence of a central SMBH of mass $M_{\bullet} \gtrsim 10^6 M_{\odot}$ would stabilise the inner region of the disc and allow growth by steady accretion as long as the gas supply remains [36]. Thus, the crucial bottleneck is the formation of the initial SMBH, such that rapid seeding or growth is unavoidable within our current understanding of the most distant quasars.

2.1. Primordial black holes

The very first black holes may be byproducts of the extreme conditions of the ultra-early Universe. Specifically, the high pressure and energy fields seeded by inhomogeneous quantum fluctuations could have triggered sufficient compression of overdense regions to give rise to primordial black holes [37]. These could be Planck relics with initial masses from $m_p \sim 10^{-8}$ kg upwards, experiencing highly inefficient growth throughout cosmic history with the possible exception of a rapid growth mode in dense galactic centres. Interestingly, miniature primordial black holes with masses $M_{\bullet} \lesssim 10^{-19} M_{\odot} \approx 10^{11}$ kg are of little astrophysical consequence as the evaporation timescale due to Hawking radiation⁶ becomes significantly less than the age of the Universe. More massive primordial black holes would persist to the present day and have been invoked as plausible dark matter candidates. However, to account for the inferred dark matter

budget, e.g. from measurements of the CMB [38], the primordial black hole mass is restricted to $20 M_{\odot} \lesssim M_{\bullet} \lesssim 100 M_{\odot}$, coincident with the masses of recently discovered coalescing binary black holes [9,39]. Lower masses are excluded by microlensing surveys [40], while high masses would disrupt wide binaries [41]. Still, primordial black holes could exist outside this range at much lower abundances, thereby providing an alternative seeding mechanism for SMBHs.

2.2. Stellar remnant black holes

The first black holes to form via conventional gravitational collapse were remnants of the first (Pop III) stars. Due to the substantially lower metal content, compared to present-day star forming clouds, these stars likely exhibited a top-heavy initial mass function [2]. The most massive stars ($\gtrsim 10 M_{\odot}$) underwent efficient nuclear fusion and had short main sequence lifetimes ($\lesssim 3$ Myr) before their eventual deaths as either core-collapse supernovae ($10-40 M_{\odot}$), superluminous pair-instability supernovae ($140-260 M_{\odot}$), or by directly collapsing to a black hole ($40-140 M_{\odot}$, $\gtrsim 260 M_{\odot}$, [42,43]). However, the first stellar remnant black holes were ‘born starving’, as their strong ionising feedback was capable of evaporating the surrounding gas [15,44]. Furthermore, without a stabilising galactic potential well they were susceptible to ejection from their host halos by natal kicks imparted during formation, precluding further growth [45,46]. The subsequent self-enrichment of metals from stochastic supernovae and external pollution from more evolved neighbouring galaxies prompted a transition to a Pop II mode of star formation with enhanced fragmentation, establishing a universal low-mass dominated initial mass function [47–49]. This marked an end to the formation of isolated massive stellar remnant black hole seeds.

2.3. Eddington-limited accretion

Once a black hole has formed it may grow by accreting gas from the surrounding medium. To be captured a particle must fall within the causal event horizon, i.e. the Schwarzschild radius $R_S = 2GM_{\bullet}/c^2$. However, under the assumption of spherical inflow all gas is susceptible to accretion if it falls within the so-called Bondi-Hoyle radius $R_B = 2GM_{\bullet}/c_s^2$, where c_s denotes the sound speed [51]. This model forms a benchmark for understanding black hole growth. The accretion rate is then regulated by the build-up of pressure from the radiation emitted by the infalling gas as it encounters viscous heating. The Eddington limit is the maximal luminosity an accreting object can

achieve before launching an intense radiation-driven wind. Under spherical accretion the classical Eddington luminosity including only electron scattering is $L_{\text{Edd}} = 4\pi GM_{\bullet} m_p c / \sigma_T \sim 10^{38} (M_{\bullet}/M_{\odot}) \text{ erg/s}$,⁷ where m_p is the proton mass and σ_T the Thomson scattering cross-section of free electrons. This limit is independent of the distance from the black hole because both the radiative and gravitational forces fall off as $1/r^2$. Furthermore, black hole energy production is highly efficient compared to other processes, e.g. nuclear fusion, with the observed luminosity conventionally given as $L = \epsilon \dot{M}_{\bullet} c^2$, where $\epsilon \sim 0.1$. This provides a constraint on SMBH growth rates, such that $\dot{M}_{\bullet} \propto M_{\bullet}$ maximally allows exponential growth.

The Eddington limit serves as the canonical criterion for inhibiting black hole growth. However, several complicating factors can alter this simple picture. For example, thin-disc configurations allow radiation to preferentially escape in the polar directions, such that the funnelled accretion process in three-dimensions remains undisturbed. The radial advection velocity in a viscous disc is given by $v_r \sim \alpha c_s H/r$, where α is the viscous parameter (< 1) and H is the disc scale height ($< r$). Thus, the high entropy gas inflows are generally subsonic $v_r \lesssim c_s$, but transport energy faster than photon-diffusion due to the high optical depths.⁸ Furthermore, other forms of radiation can provide additional feedback, porosity can generate radial stratification, and energy can be lost to the black hole itself. Fortuitous circumstances can alleviate some of the growth restrictions, and in fact a super-Eddington quasar at $z=6.6$ has recently been discovered in support of robust SMBH growth in the high-redshift Universe [52].

Still, black hole accretion in typical galactic environments is episodic as a result of self-regulating radiative feedback which yields accretion rates that are sub-Eddington when averaged over multiple duty cycles [15,16,53]. Maintaining super-Eddington accretion is possible when the black hole is embedded within sufficiently dense gas, which renders the radiation pressure less effective [54]. Analytic models using the photon trapping criterion (in which photon diffusion is slower than v_r) allow accretion levels far in excess of the Eddington rate [55]. However, such models do not account for photo-heating (hydrodynamical) feedback on the Bondi scale, which could strongly suppress the inflow rate from larger scales [56]. The necessary condition for hyper-Eddington accretion is to overcome photo-heating feedback, which can only happen if the ionised region is confined within the Bondi radius, so multi-frequency radiation hydrodynamics coupling is required. In such one- and two-dimensional radiation hydrodynamics simulations (see Figures 2 and 3), accretion

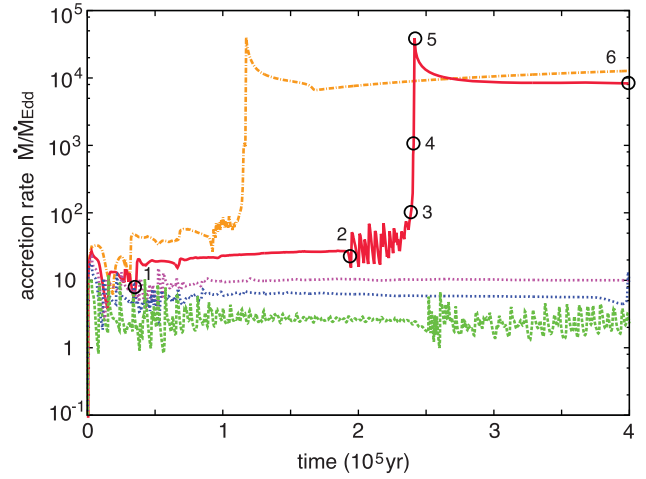


Figure 2. Accretion rate history from one-dimensional radiation-hydrodynamics simulations of black holes with mass $M_{\bullet} = \{1, 3, 5, 10, 20\} \times 10^3 M_{\odot}$ (green, blue, magenta, red, and orange, respectively). The radiative efficiency is based on the photon trapping model and the ambient gas density is $n_{\infty} = 10^5 \text{ cm}^{-3}$. The higher black hole masses ($M_{\bullet} \gtrsim 10^4 M_{\odot}$) exhibit a significant jump in their accretion rates, which maintains an approximately constant super-Eddington value as the ionised region is confined within the Bondi radius (adopted from [50]). © Kohei Inayoshi.

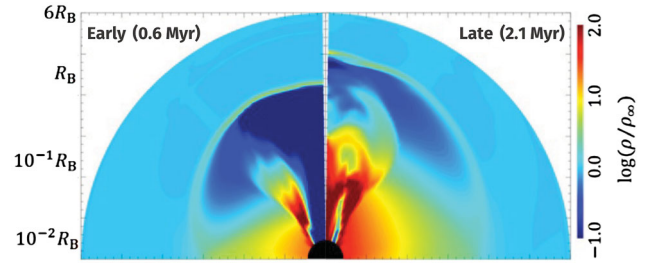


Figure 3. Two-dimensional radiation-hydrodynamics simulation demonstrating a transition to super-Eddington accretion rates. The black hole has a mass of $M_{\bullet} \approx 5 \times 10^5 M_{\odot}$ and ambient density of $n_{\infty} = 10^5 \text{ cm}^{-3}$. The left and right panels show the gas density during an early stage at $t = 0.6 \text{ Myr}$ ($\approx 0.14 t_{\text{dyn}}$) and late stage at $t = 2.1 \text{ Myr}$ ($\approx 0.49 t_{\text{dyn}}$), respectively (adapted from [58]). © Eishun Takeo.

rates can exceed $\dot{M}_{\bullet} \gtrsim 10^3 L_{\text{Edd}}/c^2$ when the following condition is satisfied: $(M_{\bullet}/10^5 M_{\odot})(n_{\infty}/10^4 \text{ cm}^{-3}) \gtrsim 2.4 (\langle \epsilon \rangle / 100 \text{ eV})^{-5/9}$, where n_{∞} is the density of the ambient gas and $\langle \epsilon \rangle$ is the mean energy of the ionising radiation from the central black hole [50,57]. It remains an open question whether super-Eddington growth rates are sustainable considering the violent assembly environments of the first galaxies where newly formed stars and supernovae blow away the surrounding gas.

2.4. Direct-collapse black holes

On the other hand, scenarios involving higher initial black hole masses offer promising alternatives to

circumvent the quasar-seed timing problem. Primordial galaxies with a delayed onset of star formation facilitate a mechanism to produce massive (10^4 – $10^6 M_\odot$) black hole seeds, which form ‘in one go’ from gas clouds with inefficient cooling. Specifically, according to the Jeans criterion for triggering gravitational instability,⁹ if such a cloud maintains high thermal pressure support then fragmentation is suppressed and the central protostellar core results in a single supermassive star as a result of the significantly altered evolutionary track through density–temperature phase space during collapse (higher thermal pressure add stability). The DCBH scenario depends heavily on the absence of molecular hydrogen, heavy elements, and dust as radiative cooling would lower the temperature and Jeans mass scale. Therefore, DCBHs can only form early in cosmic history under a specific set of rare conditions.

The canonical mechanism to destroy H_2 molecules is by photodissociation from non-ionising Lyman-Werner photons (‘LW’, with energies between 11.2–13.6 eV) emitted by the first stars [59] and DCBHs [60]. LW radiation can travel great distances and build up a LW background modulated by strong local fluctuations that entirely suppresses star formation in minihaloes with virial masses of $M_{\text{vir}} \lesssim 10^6 M_\odot$ [61–63]. More massive atomic cooling haloes, on the other hand, become self-shielding so that molecules can form again, eventually leading to star formation [64]. This means that self-consistent calculations must consider both the radiative transfer and chemical rate equations [65–69]. Furthermore, the shape of the spectral energy distribution affects the relevant photodissociation and photodetachment rates so that it is technically insufficient to parametrise outcomes based only on the strength of the LW radiation field required to suppress H_2 cooling, typically $J_{\text{crit}} \sim 10^2$ – 10^4 in conventional units of 10^{-21} erg/s/cm²Hz/sr [62,70]. Nonetheless, the first simulations with cosmological initial conditions presumed prior destruction of H_2 via a super-critical LW flux, demonstrating that the inflow would continue on its near-isothermal track [18]. The resulting free-fall collapse of the atomically cooling gas could then produce massive black holes directly. Subsequently, the DCBH model has received increased attention with a particular focus on the conditions of collapse, overcoming the angular momentum barrier, consequences of disc (in)stability, supersonic turbulence properties, formation of a radiation pressure-supported supermassive star, accretion rates, feedback processes, and final DCBH mass function and abundance estimates in the cosmological context [71–75].

Semi-analytic models and hydrodynamics simulations suggest that strong LW radiation fields can be supplied directly from neighbouring galaxies in close proximity

to the DCBH formation site [76–79]. This scenario is most plausible for a pair of coincidentally synchronised galaxies in a ‘Goldilocks zone’ of separation and timing, to maintain sufficient flux while avoiding ram-pressure stripping or external metal pollution. However, statistical searches for neighbouring halo DCBH candidates in large-volume cosmological simulations find much fewer than expected [80,81], and therefore only represents a fraction of the present-day SMBH population. Still, there are several other mechanisms to enable more robust DCBH formation under less extreme LW radiation conditions. For example, it has been suggested that trapped $\text{Ly}\alpha$ cooling radiation during the initial collapse may enhance the assembly of DCBHs by facilitating the photodetachment of H^- ions, which are chemical precursors to H_2 [70,82], or by stiffening the thermodynamic equation of state for the gas directly [83,84]. Alternatively, strongly amplified magnetic fields and turbulence can provide additional pressure support on small scales to increase the viability of the DCBH scenario [85].

More generally, shock heating due to supersonic accretion along cosmological filaments, high baryonic streaming velocities relative to dark matter, or violent merger histories may suppress star formation in route to DCBH formation [87,88]. In particular, streaming velocities have been shown to play an important role in suppressing star formation in minihaloes [89–92], and may be a powerful mechanism to enhance the formation of massive seed black holes [93]. Numerical models may include any of these effects from the initial collapse through the accretion and feedback phases with high accuracy [94]. However, fully coupled, radiation hydrodynamics simulations hold the key for further progress and already indicate profound differences from an adiabatic equation of state during both early and late stages affecting the growth of the central structure and anisotropic escape of ionising photons in 3D geometry [95,96]. By utilising such simulations it was recently argued that the dynamics of structure formation, rather than a critical LW flux, is the main driver of the formation of DCBHs [97]. This paradigm, or similar variations [98], is attractive because massive black hole seeds can form with less stringent metallicity requirements, implying that they may be much more common than previously considered in overdense regions of the early Universe.

2.5. Runaway mergers in dense star clusters

The final SMBH formation scenario to be discussed is that of dense clusters undergoing runaway collapse. Such an origin represents an intermediate case between monolithic collapse of DCBHs and the staggering growth required for isolated stellar remnant black holes. In fact,

At the extreme end of SMBH-accretion powered AGN activity are the quasars, with distinguishing characteristics, as follows: (i) the luminosity can be orders of magnitude brighter than their host galaxy, (ii) the central engines are extremely small and unresolved in observations, (iii) the significant variability of the observed flux over short timescales, and (iv) the non-thermal spectral energy distributions, which often follow a power law out to high energies with strong X-ray emission. The

non-thermal continuum is generated by bremsstrahlung radiation due to the acceleration of electrons by ions, Compton scattering of photons by free electrons with the corresponding frequency shifting, and synchrotron radiation due to electrons accelerating in a magnetic field. Additionally, the accretion disc radiates broadband emission with a peak in the UV, corresponding to a characteristic blackbody temperature for the Eddington luminosity near the event horizon of $T_{\text{Edd}} \approx 5 \times 10^5 \text{ K} (M_{\bullet}/10^8 M_{\odot})^{-1/4}$ [17]. The canonical AGN morphology is that of an obscuring molecular torus surrounding a compact accretion disc, with fast moving ($\gtrsim 1000 \text{ km/s}$) broad-line-region clouds out to tens of parsecs transitioning to narrow-line-region clouds at larger distances. The relative importance of these features including the presence of a relativistic jet along with the observed orientation provides a unification scheme for AGN classifications [113].

Cosmological hydrodynamics simulations often do not resolve the relevant scales for SMBH growth. The approximate models are based on inefficient thermal feedback at high accretion rates as the dominant growth channel, until a self-regulating kinetic feedback mode maintains lower accretion rates with SMBH mergers becoming the main channel for subsequent mass growth [114,115]. In the first galaxies, radiation pressure from the black hole along with concurrent star formation and supernovae are likely to have had a substantial impact on their surrounding environments [7]. Lower mass minihaloes with shallower gravitational potential wells ($M_{\text{vir}} \sim 10^5\text{--}10^7 M_{\odot}$) would have been especially susceptible to radiative feedback, which potentially depletes the reservoir of gas needed to fuel black hole growth [116]. A particularly interesting source of feedback arises from the resonant scattering of Ly α photons (see Figure 5). In 1D simulations, a radial outflow forms in response to the central source ionising and heating the gas, with Ly α photons significantly enhancing such radiation-driven winds [117,118]. An approximate model for Ly α pressure was recently developed for 3D hydrodynamics simulations [119], which demonstrated an impact on metal-poor dwarf galaxies by regulating the dynamics of star-forming clouds before the onset of supernova explosions. Future cosmological simulations with an accurate and efficient treatment of Ly α feedback will provide new perspectives on the growth of the first stars and SMBHs, cold gas accretion flows, and turbulence in high-redshift galaxies [120,121].

3.2. Co-evolution with stars

The existence of SMBHs at the centre of almost every galaxy provides strong evidence for a generic growth

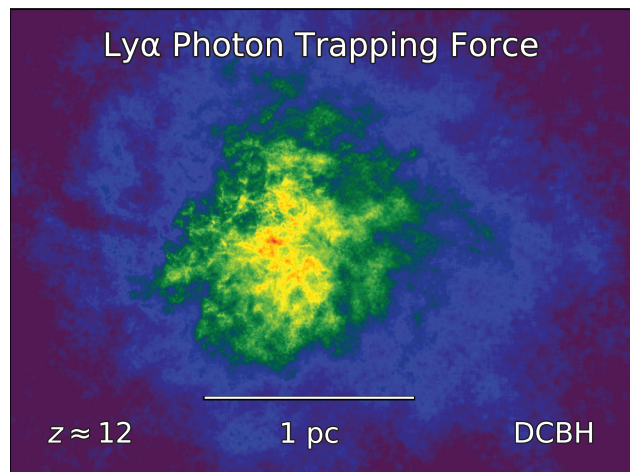


Figure 5. Illustration of Ly α photon trapping in a DCBH assembly environment. The central regions (red–green) experience strong turbulent accelerations of order $a_{\text{Ly}\alpha} \gtrsim 10 \text{ km/s/kyr}$ due to Ly α radiative feedback. Although this problem requires fully-coupled radiation hydrodynamics, post-processing radiative transfer simulations already indicate that Ly α radiation pressure can be dynamically important to first aid DCBH formation and then provide self-regulating feedback to limit the final mass (adapted from [122]). © Aaron Smith.

mechanism alongside that of the host galaxy. Indeed, a remarkable correlation has been established between the central black hole mass $M_{\bullet}$ and stellar velocity dispersion $\sigma_{\star}$ in the bulge (central nucleus) of galaxies: $M_{\bullet} = 0.310^{+0.037}_{-0.033} \times 10^9 M_{\odot} [\sigma_{\star}/(200 \text{ km/s})]^{4.38 \pm 0.29}$ (see Figure 6, [123–125]). The so-called $M_{\bullet}$ – $\sigma_{\star}$ relation strongly suggests the existence of a feedback mechanism to couple the growth of nuclear SMBHs to the mass of the galaxy. In fact, the co-evolution could originate as a consequence of the SMBH formation mechanism. For example, the collapse of primordial gas clouds with successful massive seed black holes would be followed by star formation in the spheroidal bulge [60]. The virialization balance between the gravitational potential and kinetic energies produce the observed stellar velocity dispersion. As black hole accretion continues, the intense radiative feedback periodically regulates the gas supply to the bulge and the feeding of the black hole. On the other hand, SMBHs do not correlate with galaxy-scale discs and are only weakly related to disc-grown ‘pseudobulges’ or properties of the host dark matter haloes. This suggests multiple regimes of black hole feedback including effects from the local environment and galaxy mergers that complicate the high-level picture described above.

Another aspect of co-evolution is the existence of nuclear star clusters and nuclear discs that co-exist with SMBHs at the centres of low- and intermediate-mass spheroids [126]. Interestingly, massive elliptical galaxies

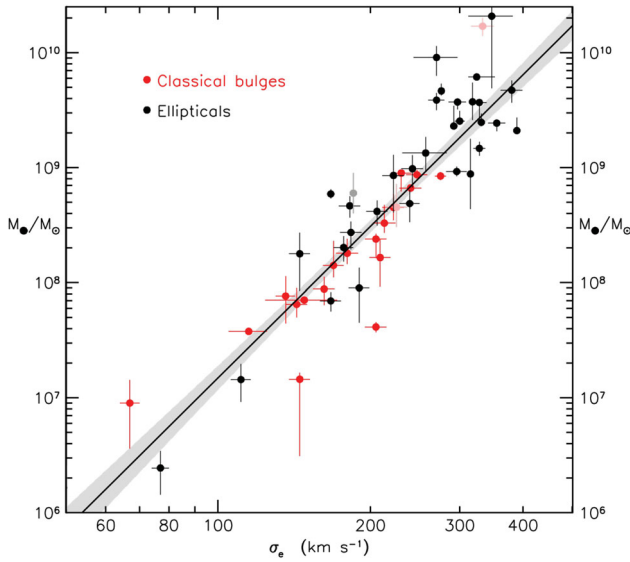


Figure 6. The $M_{\bullet}-\sigma_{*}$ relation is based on the remarkable correlation between the observed central black hole mass $M_{\bullet}$ and stellar velocity dispersion σ_{*} in the bulge of galaxies, suggesting the existence of a feedback mechanism to couple the growth of nuclear SMBHs to the mass of the galaxy. The relation extends across many orders of magnitude in mass and holds for both classical bulges and elliptical galaxies (morphological classifications). The line is a symmetric least-squares fit to the data with the 1σ range shown as shaded regions. (adapted from [123]). © John Kormendy.

do not contain nuclear star clusters, presumably as a result of dynamical friction and evaporation during hierarchical merging. It is also possible that the most extreme SMBH formation sites were violent enough to also preclude efficient nuclear star formation. However, at decreasing spheroid mass, the nuclear star cluster eventually dominates over the SMBH component, consistent with an absence of a central black hole in globular clusters [127]. When both a SMBH and nuclear star cluster exist, it may make sense to combine the masses for an $(M_{\bullet} + M_{\text{NSC}})-\sigma_{*}$ relation with less intrinsic scatter [128,129]. Inferences about high-redshift low-luminosity AGN can be affected by the presence of nuclear star clusters due to competing feedback, biased velocity dispersions, and errors in mass estimates, while at the same time potentially providing clues about the evolution of the first SMBHs. Recent observations indicate that the majority of $z \sim 6$ quasars are more massive than expected from the local $M_{\bullet}-M_{\text{halo}}$ relation, with a third of the known population by factors $\gtrsim 10^2$ [130]. These numbers are likely affected by selection bias and unknown completeness, but clearly indicate that the growth of the most massive high-redshift SMBHs greatly precedes that of hosting haloes, implying that the $M_{\bullet}-\sigma_{*}$ relation is not initially established via symbiotic growth at early times [28].

3.3. Mini-quasars

Radiation from the first stars and galaxies initiated the dramatic phase transition known as the ‘epoch of reionization’ marking an end to the cosmic dark ages within the first billion years after the Big Bang. While it is generally accepted that stars contributed a significant fraction of the ionising photon budget, the role of X-ray sources including low-luminosity mini-quasars to the reionization process is tentatively constrained to be subdominant [131,132]. The X-rays from early AGN heat the IGM above the CMB temperature, rendering the 21 cm hyperfine-structure line of hydrogen visible in emission at $z \lesssim 15$ [22,23]. Both the neutral hydrogen fraction and 21 cm spin temperature depend on the spectral shape of the ionising radiation, giving rise to either a more patchy or homogeneous IGM, with soft X-ray sources being more efficient at heating and ionising than hard X-ray sources [133]. There seems to be mounting evidence for so-called ‘late’ reionization in which most of the volume of the Universe is rapidly ionised around $z \sim 7-8$ as opposed to the previous idea that reionization was well under way by $z \sim 10$. This understanding comes from: (i) the *Planck* measurement of a low optical depth for electron scattering of CMB photons ($\tau_e = 0.054 \pm 0.007$, [134]), (ii) the declining fraction of Ly α emitters among the galaxy population at $z \gtrsim 6$ [135–137], (iii) the imprint of neutral hydrogen in the IGM as a damping wing absorption feature on the spectrum of high-redshift quasars [138,139], and (iv) the thermal history of the IGM [140,141]. Models with dominant AGN contributions are typically disfavoured due to constraints of the quasar luminosity function at $z \lesssim 5$ and delayed helium reionization at $z \lesssim 3$ [142,143]. On the other hand, Ly α forest observations also provide a measurement of the integrated time that galaxies shine as AGN. Such a procedure has identified a population of high-redshift quasars with very small proximity zones, indicating active lifetimes of less than 10^4 years in support of massive seeding and rapid growth scenarios [144,145]. We anticipate a more complete understanding of the epoch of reionization from complementary upcoming facilities, including the SKA and JWST, which will provide essential clues about the formation and evolution of SMBHs in the high-redshift Universe.

4. Detection

Our emerging understanding of structure formation, within the overall framework of standard cosmology, clearly indicates that black holes play an important role already early on in cosmic history. It is therefore important to develop a network of empirical probes to test

the theoretical picture. There are a number of promising avenues to accomplish this in the upcoming decade and beyond with a suite of next-generation observational facilities. The upcoming surveys will cover a broad range of wavelengths, and crucially will have a multi-messenger character. In briefly summarising some of those approaches, it is useful to distinguish direct (*in-situ*) observations at high redshifts from ‘fossil’ probes, where signatures of the first black holes are captured at later cosmic times.

4.1. In-situ methods

A key target for high-redshift observations is to look for evidence of DCBH seeds. Since these objects are ‘born massive’, they would stand out in a number of ways. First, they would initially present a pure black hole (AGN) signature, with a weak, or non-existent, stellar component in the overall source spectral energy distribution (SED). Such an early AGN-dominated phase has been termed ‘obese’, and has been suggested as a telltale DCBH feature [146, see Figure 7]. Second, the early growth phase of a DCBH is predicted to be heavily obscured, where a massive gaseous envelope is rapidly accreted onto the central source [147,148]. The high column density in neutral hydrogen will efficiently absorb UV photons, produced in the inner accretion flow, above the hydrogen-ionisation threshold. Those photons will be reprocessed into lower-energy infrared radiation, whereas any non-thermal X-ray photons can largely

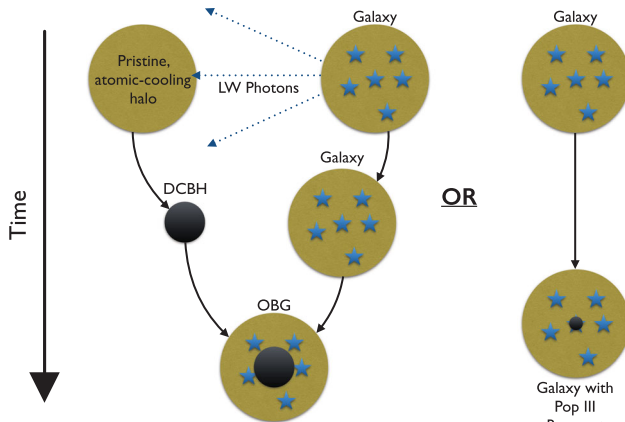


Figure 7. Schematic illustration of two seed scenarios in which the black hole is either obese or subdominant. *Left panel:* A DCBH seed ($\sim 10^5 M_\odot$) merges with its parent star-forming halo, which provided the LW radiation that facilitated its assembly. *Right panel:* A Pop III stellar remnant black hole ($\sim 100 M_\odot$) grows within a previously star-forming galaxy. Multi-wavelength observations and predictions will play an important role in distinguishing formation scenarios in the era of next-generation telescopes, such as the *JWST* (adopted from [146]). © Priyamvada Natarajan.

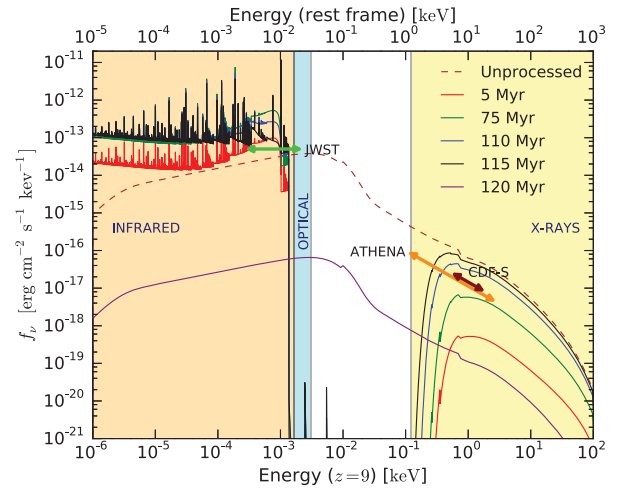


Figure 8. Emerging spectrum from a DCBH host system at $z \sim 10$. It is evident how the unprocessed spectrum (*dashed line*) is reprocessed by the high column-density in the envelope into the characteristic ‘double-humped’ shape during early DCBH evolution. Eventually, after ~ 120 Myr, the previous accretion has depleted the high-density gas, so that the central spectral gap is filled in again. The key wavebands for capturing this spectral signature are the near-infrared with the *JWST*, and the X-ray band with *ATHENA* (sensitivity thresholds are indicated) (adopted from [148]). © Fabio Pacucci.

escape unobscured. The resulting radiative transport thus imprints a characteristic ‘double-humped’ spectral profile, with a peak in the near- and mid-infrared, as well as $\sim$ keV X-ray bands (see Figure 8). Deeply embedded inside the massive accretion flow, an initially ultra-compact ionised region is developing [122]. At some stage, this growing ionised region will break out of the embedding cocoon, thus effectively terminating the rapid DCBH assembly process. As a consequence of the dynamics, photons will experience an increased opacity to scattering from the abundant free electrons. The flow, in effect, becomes ‘Compton thick’, or more precisely ‘Thomson thick’, such that $\tau_e \gtrsim 1$ (see footnote 8). After a few 10 Myr, this initial, highly obscured and possibly Compton-thick, evolutionary phase will end, and the emerging radiation will transition into a standard AGN signature. The precise duration of this initial stage is uncertain, thus impacting any assessment for the likelihood of catching a DCBH in its early characteristic phase.

To hunt for the DCBH signature, *JWST* observations in the near- and mid-infrared will have the required sensitivity for broad-band photometry with its NIR-Cam and MIRI instruments. Given the uncertainty in predicting source number densities, such deep *JWST* searches are ideally complemented by wide-field surveys with the *Wide Field Infrared Survey Telescope* (*WFIRST*),

further ahead in time. Similarly, it will be crucial to mirror the infrared observations with simultaneous X-ray campaigns. The latter are ideal targets for the planned *Advanced Telescope for High ENergy Astrophysics* (ATHENA) or *Lynx* X-ray missions. In terms of such correlated signatures in the infrared and X-ray bands, there is an independent strategy to access it, by focussing on the aggregate signal imprinted in the cosmic infrared and X-ray backgrounds [149]. Indeed, there are tantalising hints for a correlated signal in the two backgrounds, with statistical properties that can be attributed to an unresolved population of high-redshift sources that emit *both* in soft-UV photons and hard X-rays. After cosmological redshifting, those emissions would be deposited in the present-day cosmic backgrounds [25]. It is possible that DCBH sources are, at least partially, responsible for the inferred signature in these cosmic backgrounds [150].

4.2. Fossil methods

The gravitational-wave signature from merging black hole binaries offers an exciting new window into the cosmic history of black hole formation and growth. With advanced LIGO, the black hole remnants from massive Pop III progenitor stars may be detectable as a subset of the rapidly growing overall sample [26,153,154]. Any predictions for the expected Pop III binary black hole merger rates as a function of redshift are subject to uncertainties in the input physics, such as the Pop III initial mass function and binary stellar evolution at zero metallicity [155]. Approximate calculations, however, indicate that of order 1% of the total LIGO binary black hole sample may originate in Pop III stellar systems, with about one expected merger per year at LIGO's final design sensitivity (see Figure 9). A key challenge is to disentangle any Pop III gravitational-wave signal from the dominant population of metal-enriched progenitors, given that the currently observed black hole mass range of a few tens of solar masses in itself is not uniquely identifying Pop III. Such a unique Pop III signature would be provided by black hole masses of $\sim 300 M_{\odot}$, given that metal-enriched massive stars are not likely to reach this extreme mass range. In the latter case, vigorous stellar winds would trigger strong mass loss, which may largely be absent for Pop III stars.

To reach the supermassive regime with gravitational-wave observations, we will have to wait for the Laser Interferometer Space Antenna (LISA) to fly in the 2030s. This space-based observatory with its million-km baselines is sensitive to frequencies of ~ 0.1 mHz– 0.1 Hz, thus extending the LIGO band to much longer wavelengths, where the signature of the first SMBHs is expected [156]. Specifically, the LISA sensitivity should

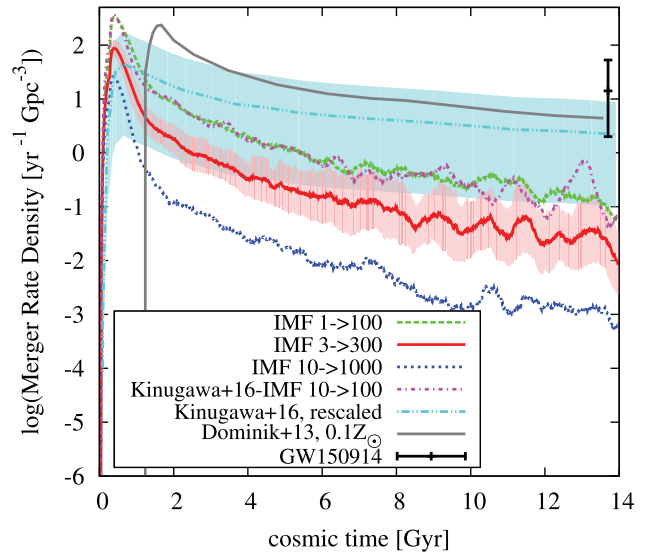


Figure 9. Compact binary merger rate for Pop III sources, plotted as a function of cosmic time, for different assumptions on the initial mass function. The thick red line with a shaded error region represents the preferred model. Similar curves from other studies are reproduced for reference [151,152]. The black data point with error bars (*upper right*) is the estimated binary black hole merger rate as derived from the initial LIGO detection. As can be seen, Pop III sources may contribute of order 1% of the full LIGO sample (adopted from [26]). © Tilman Hartwig and Volker Bromm.

enable the detection of black hole mergers with masses $\sim 10^3$ – $10^6 M_{\odot}$, all the way to redshifts of $z \sim 20$. One of the key questions now is how to distinguish between the two main scenarios to seed black hole growth, light stellar seeds and heavy DCBH ones [157]. Through a combination of X-ray observations at high-redshift (see above) and the statistical analysis of the gravitational-wave events, as detected by LIGO and LISA, we can build up the mass spectrum of the first black holes. In principle, this determination should allow us to identify the dominant black hole formation channel in the pre-reionization Universe.

5. Outlook

Our emerging picture of early cosmological structure formation contains the intriguing message that black holes played an important role early on. Indeed, the first black holes provide us with a multitude of probes into the formative initial period of cosmic history. Furthermore, it has become evident that they were key players, actively shaping early cosmic evolution through their feedback effects on the primordial gas. This feedback in turn comprises a large range of scales, from the immediate surroundings of the compact object, to the interstellar medium in the host haloes, reaching all the way into the large-scale cosmic web. The Universe throughout cosmic

times contains black holes over a wide range of masses, but it is remarkable that black hole formation in the early Universe may have received a ‘headstart’, biasing any formation channels to higher masses. Such possible bias would mirror the predicted top-heavy nature of the first (Pop III) stars.

The first black holes also add powerful new avenues for observational astronomy. At the supermassive scale, the *JWST* may detect the initial, heavily-obscured, formation phase of DCBHs, as well as their subsequent accretion and growth phase. Crucially, these sources provide us with portals into multi-wavelength and multi-messenger surveys. As to the former, the spectral energy distribution of the black hole emission is extended into the X-ray regime, opening up the high-energy window both for individual sources and their combined contribution to the cosmic X-ray background. At the extreme energy end, a subset of the first black holes is expected to be born in rare gamma-ray burst events [158], accessible to future all-sky gamma-ray burst missions, such as the Transient High-Energy Sky and Early Universe Surveyor (THESEUS). Additionally, 21 cm cosmology measurements with the SKA will provide key insights via constraints on the cold primordial gas in the cosmic dark ages and the epoch of reionization. Even more exciting, as part of the developing field of multi-messenger astronomy, are the prospects to link the electromagnetic emission from the first black holes with their gravitational-wave signature, and possibly even with the search for ultra-energetic neutrinos. Given the confluence of breakthroughs in the theoretical modelling as well as in observational capabilities, the upcoming decade promises to be a golden age of discovery for the first supermassive objects in the Universe.

Notes

1. The first stars, formed out of primordial hydrogen and helium gas, are often called Population III (Pop III) stars. This terminology extends the traditional sequence of astronomy, where metal-rich stars, like our Sun, are Population I stars, and the metal-poor stars in the outer regions of the Milky Way constitute Population II.
2. In astronomy, the mass of the Sun is denoted with $M_{\odot} \approx 2 \times 10^{30}$ kg.
3. Intuitively, m_p is the mass of a black hole that is just massive enough to gravitationally hold on to its mass, before quantum-tunnelling is able to return matter to the outside of the black hole event horizon. Classically, nothing is allowed to do that.
4. Standard analysis of the primordial chemistry indicates that the dark matter has to be ‘non-baryonic’, made up of a new kind of elementary particle.
5. These photons reside in the so-called Lyman-Werner (LW) wavelength band, giving rise to the term LW radiation.

6. Hawking radiation, named after the British theoretical physicist Stephen Hawking, is black-body radiation predicted to be released by black holes due to quantum effects near the event horizon.
7. Astronomers often use ‘erg’ for units of energy, which translates to SI units as $1 \text{ erg/s} = 10^{-7} \text{ W}$.
8. ‘Optical depth’ is a term that describes the (logarithmic) transmission of light through a material, and is calculated along ray segments as $\tau_e = \int_0^L n_e \sigma_T d\ell$, where n_e denotes the number of free electrons per unit volume.
9. The Jeans instability causes gas clouds to compress and fragment when the internal pressure is not strong enough to prevent gravitational collapse. This occurs when the free-fall time, $t_{\text{ff}} \approx 1/\sqrt{G\rho}$, is shorter than the sound-crossing time, $t_s \approx R/c_s$, with G the universal gravitational constant, ρ the density, R the radius, and c_s the sound speed.

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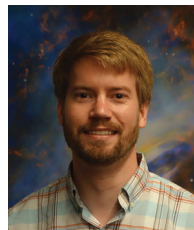
Disclosure statement

No potential conflict of interest was reported by the authors.

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References

- [1] Chandrasekhar S. The black hole in astrophysics: the origin of the concept and its role. *Contemp Phys.* **1974 Jan**;15: 1–24.
- [2] Bromm V, Yoshida N. The first galaxies. *ARA&A.* **2011 Sep**;49: 373–407.
- [3] Loeb A, Furlanetto SR. The first galaxies in the universe. Princeton, NJ: Princeton Univ. Press; **2013**.
- [4] Glover S. The first stars. In: Wiklind T, Mobasher B, Bromm V, editors. The first galaxies. *Astrophysics and space science library*, vol. 396. Berlin, Heidelberg: Springer; **2013**.
- [5] Barger AJ. The history of star formation in the universe. *Contemp Phys.* **2017 Apr**;58(2):160–175.
- [6] Madau P, Rees MJ. Massive black holes as population III remnants. *ApJ* **Apr.** **2001**;551: L27–L30.
- [7] Jeon M, Pawlik AH, Greif TH, et al. The first galaxies: assembly with black hole feedback. *ApJ.* **2012 Jul**;754: 34.
- [8] Jeon M, Pawlik AH, Bromm V, et al. Radiative feedback from high-mass X-ray binaries on the formation of the first galaxies and early reionization. *MNRAS.* **2014 Jun**;440: 3778–3796.
- [9] Bird S, Cholis I, Muñoz JB, et al. Did LIGO detect dark matter? *Phys Rev Lett.* **2016 May**;116:201301.
- [10] Fan X, Strauss MA, Richards GT, et al. A survey of $z > 5.7$ quasars in the sloan digital sky survey. IV. Discovery of seven additional quasars. *AJ.* **2006 Mar**;131: 1203–1209.
- [11] Mortlock DJ, Warren SJ, Venemans BP, et al. A luminous quasar at a redshift of $z = 7.085$. *Nature.* **2011 Jun**;474: 616–619.
- [12] Wu XB, Wang F, Fan X, et al. An ultraluminous quasar with a twelve-billion-solar-mass black hole at redshift 6.30. *Nature.* **2015 Feb**;518: 512–515.
- [13] Bañados E, Venemans BP, Mazzucchelli C, et al. An 800-million-solar-mass black hole in a significantly neutral Universe at a redshift of 7.5. *Nature.* **2018, Jan**;553: 473–476.
- [14] Haiman Z, Loeb A. What is the highest plausible redshift of luminous quasars? *ApJ.* **2001 May**;552: 459–463.
- [15] Johnson JL, Bromm V. The aftermath of the first stars: massive black holes. *MNRAS.* **2007 Feb**;374: 1557–1568.
- [16] Milosavljević M, Bromm V, Couch SM, et al. Accretion onto ‘seed’ black holes in the first galaxies. *ApJ.* **2009 Jun**;698: 766–780.
- [17] Rees MJ. Black hole models for active galactic nuclei. *ARA&A.* **1984 Jan**;22: 471–506.
- [18] Bromm V, Loeb A. Formation of the first supermassive black holes. *ApJ.* **2003 Oct**;596: 34–46.
- [19] Mirabel IF, Dijkstra M, Laurent P, et al. Stellar black holes at the dawn of the universe. *A&A.* **2011 Apr**;528: A149.
- [20] Kuhlen M, Madau P. The first miniquasar. *MNRAS.* **2005 Nov**;363: 1069–1082.
- [21] Galli D, Palla F. The dawn of chemistry. *ARA&A.* **2013 Aug**;51: 163–206.
- [22] Furlanetto SR, Oh SP, Briggs FH. Cosmology at low frequencies: the 21 cm transition and the high-redshift Universe. *Phys Rep.* **2006 Oct**;433: 181–301.
- [23] Pritchard JR, Loeb A. 21 cm cosmology in the 21st century. *Rep Prog Phys.* **2012 Aug**;75:086901.
- [24] Bowman JD, Rogers AEE, Monsalve RA, et al. An absorption profile centred at 78 megahertz in the sky-averaged spectrum. *Nature.* **2018 Mar**;555: 67–70.
- [25] Cappelluti N, Kashlinsky A, Arendt RG, et al. Cross-correlating cosmic infrared and X-ray background fluctuations: evidence of significant black hole populations among the CIB sources. *ApJ.* **2013 May**;769: 68.
- [26] Hartwig T, Volonteri M, Bromm V, et al. Gravitational waves from the remnants of the first stars. *MNRAS.* **2016 Jul**;460: L74–L78.
- [27] Woods TE, Agarwal B, Bromm V, et al. Titans of the early universe: the prato statement on the origin of the first supermassive black holes. *arXiv e-prints.* **2018 Oct.** arXiv:1810.12310.
- [28] Volonteri M. The formation and evolution of massive black holes. *Science.* **2012 Aug**;337: 544.
- [29] Haiman Z. The formation of the first massive black holes. In: Wiklind T., Mobasher B., Bromm V, editors. The first galaxies. *Astrophysics and space science library*, vol. 396. Berlin, Heidelberg: Springer; **2013**.
- [30] Johnson JL, Haardt F. The early growth of the first black holes. *PASA.* **2016 Mar**;33: e007.
- [31] Latif MA, Ferrara A. Formation of supermassive black hole seeds. *PASA.* **2016 Oct**;33: e051.
- [32] Smith A, Bromm V, Loeb A. The first supermassive black holes. *Astron Geophys.* **2017 Jun**;58: 3.22–3.26.
- [33] Loeb A, Rasio FA. Collapse of primordial gas clouds and the formation of quasar black holes. *ApJ.* **1994 Sep**;432: 52.
- [34] Eisenstein DJ, Loeb A. Origin of quasar progenitors from the collapse of low-spin cosmological perturbations. *ApJ.* **1995 Apr**;443: 11.
- [35] Sugimura K, Hosokawa T, Yajima H, et al. Stunted accretion growth of black holes by combined effect of the flow angular momentum and radiation feedback. *MNRAS.* **2018 Aug**;478: 3961–3975.
- [36] Li Y, Hernquist L, Robertson B, et al. Formation of $z \sim 6$ quasars from hierarchical galaxy mergers. *ApJ.* **2007 Aug**;665: 187–208.
- [37] Carr BJ, Hawking SW. Black holes in the early universe. *MNRAS.* **1974 Aug**;168: 399–416.
- [38] Planck Collaboration, Ade PAR, Aghanim N, et al. Planck 2015 results. XIII. Cosmological parameters. *A&A.* **2016 Sep**;594: A13.
- [39] Abbott BP, Abbott R, Abbott TD, et al. Observation of gravitational waves from a binary black hole merger. *Phys Rev Lett.* **2016 Feb**;116:061102.
- [40] Alcock C, Allsman RA, Alves DR, et al. MACHO project limits on black hole dark matter in the $1\text{--}30 M_{\text{solar}}$ range. *ApJ.* **2001 Apr**;550: L169–L172.
- [41] Yoo J, Chanamé J, Gould A. The end of the MACHO era: limits on halo dark matter from stellar halo wide binaries. *ApJ.* **2004 Jan**;601: 311–318.
- [42] Heger A, Woosley SE. The nucleosynthetic signature of population III. *ApJ.* **2002 Mar**;567: 532–543.
- [43] Heger A, Fryer CL, Woosley SE, et al. How massive single stars end their life. *ApJ.* **2003 Jul**;591: 288–300.
- [44] Whalen D, Abel T, Norman ML. Radiation hydrodynamic evolution of primordial H II regions. *ApJ.* **2004 Jul**;610: 14–22.
- [45] Whalen DJ, Fryer CL. The formation of supermassive black holes from low-mass pop III seeds. *ApJ.* **2012 Sep**;756: L19.
- [46] Smith BD, Regan JA, Downes TP, et al. The growth of black holes from population III remnants in the

- renaissance simulations. *MNRAS*. 2018 Nov;480: 3762–3773.
- [47] Safranek-Shrader C, Milosavljevic M, Bromm V. Formation of the first low-mass stars from cosmological initial conditions. *MNRAS*. 2014 May;440: L76–L80.
- [48] Smith BD, Wise JH, O’Shea BW, et al. The first population II stars formed in externally enriched mini-haloes. *MNRAS*. 2015 Sep;452: 2822–2836.
- [49] Jeon M, Besla G, Bromm V. Connecting the first galaxies with ultrafaint dwarfs in the local group: chemical signatures of population III stars. *ApJ*. 2017 Oct;848: 85.
- [50] Inayoshi K, Haiman Z, Ostriker JP. Hyper-Eddington accretion flows on to massive black holes. *MNRAS*. 2016 Jul;459: 3738–3755.
- [51] Bondi H. On spherically symmetrical accretion. *MNRAS*. 1952 Jan;112: 195.
- [52] Tang JJ, Goto T, Ohya Y, et al. Rapid black hole growth at the dawn of the universe: a super-Eddington quasar at $z = 6.6$. *MNRAS*. 2019 Apr;484: 2575–2586.
- [53] Park K, Ricotti M. Accretion onto intermediate-mass black holes regulated by radiative feedback. I. Parametric study for spherically symmetric accretion. *ApJ*. 2011 Sep;739: 2.
- [54] Begelman MC. Can a spherically accreting black hole radiate very near the Eddington limit? *MNRAS*. 1979 Apr;187: 237–251.
- [55] Wyithe JSB, Loeb A. Photon trapping enables super-Eddington growth of black hole seeds in galaxies at high redshift. *MNRAS*. 2012 Oct;425: 2892–2902.
- [56] Pacucci F, Volonteri M, Ferrara A. The growth efficiency of high-redshift black holes. *MNRAS*. 2015 Sep;452: 1922–1933.
- [57] Takeo E, Inayoshi K, Ohsuga K, et al. Super-Eddington growth of black holes in the early Universe: effects of disk radiation spectra. *arXiv e-prints*. 2019 Jan. arXiv:1901.04514.
- [58] Takeo E, Inayoshi K, Ohsuga K, et al. Rapid growth of black holes accompanied with hot or warm outflows exposed to anisotropic super-Eddington radiation. *MNRAS*. 2018 May;476: 673–682.
- [59] Schaerer D. On the properties of massive Population III stars and metal-free stellar populations. *A&A*. 2002 Jan;382: 28–42.
- [60] Barrow KSS, Aykutalp A, Wise JH. Observational signatures of massive black hole formation in the early Universe. *Nat Astron*. 2018 Sep;2: 987–994.
- [61] Machacek ME, Bryan GL, Abel T. Simulations of pregalactic structure formation with radiative feedback. *ApJ*. 2001 Feb;548: 509–521.
- [62] Omukai K. Primordial star formation under far-ultraviolet radiation. *ApJ*. 2001 Jan;546: 635–651.
- [63] Agarwal B, Khochfar S, Johnson JL, et al. Ubiquitous seeding of supermassive black holes by direct collapse. *MNRAS*. 2012 Oct;425: 2854–2871.
- [64] Oh SP, Haiman Z. Second-generation objects in the universe: radiative cooling and collapse of halos with virial temperatures above 10^4 K. *ApJ*. 2002 Apr;569: 558–572.
- [65] Draine BT, Bertoldi F. Structure of stationary photodissociation fronts. *ApJ*. 1996 Sep;468: 269.
- [66] Kitayama T, Yoshida N, Susa H, et al. The structure and evolution of early cosmological H II regions. *ApJ*. 2004 Oct;613: 631–645.
- [67] Wolcott-Green J, Haiman Z, Bryan GL. Photodissociation of H_2 in protogalaxies: modelling self-shielding in three-dimensional simulations. *MNRAS*. 2011 Dec;418: 838–852.
- [68] Schauer ATP, Whalen DJ, Glover SCO, et al. Lyman-Werner UV escape fractions from primordial haloes. *MNRAS*. 2015 Dec;454: 2441–2450.
- [69] Schauer ATP, Agarwal B, Glover SCO, et al. Lyman-Werner escape fractions from the first galaxies. *MNRAS*. 2017 May;467: 2288–2300.
- [70] Agarwal B, Khochfar S. Revised rate coefficients for H_2 and H^+ destruction by realistic stellar spectra. *MNRAS*. 2015 Jan;446: 160–168.
- [71] Begelman MC, Volonteri M, Rees MJ. Formation of supermassive black holes by direct collapse in pregalactic haloes. *MNRAS*. 2006 Jul;370: 289–298.
- [72] Lodato G, Natarajan P. Supermassive black hole formation during the assembly of pre-galactic discs. *MNRAS*. 2006 Oct;371: 1813–1823.
- [73] Lodato G, Natarajan P. The mass function of high-redshift seed black holes. *MNRAS*. 2007 May;377: L64–L68.
- [74] Regan JA, Haehnelt MG. Pathways to massive black holes and compact star clusters in pre-galactic dark matter haloes with virial temperatures $\gtrsim 10000$ K. *MNRAS*. 2009 Jun;396: 343–353.
- [75] Choi JH, Shlosman I, Begelman MC. Supermassive black hole formation at high redshifts via direct collapse: physical processes in the early stage. *ApJ*. 2013 Sep;774: 149.
- [76] Dijkstra M, Haiman Z, Mesinger A, et al. Fluctuations in the high-redshift Lyman-Werner background: close halo pairs as the origin of supermassive black holes. *MNRAS*. 2008 Dec;391: 1961–1972.
- [77] Agarwal B, Dalla Vecchia C, Johnson JL, et al. The First Billion Years project: birthplaces of direct collapse black holes. *MNRAS*. 2014 Sep;443: 648–657.
- [78] Visbal E, Haiman Z, Bryan GL. Direct collapse black hole formation from synchronized pairs of atomic cooling haloes. *MNRAS*. 2014 Nov;445: 1056–1063.
- [79] Regan JA, Visbal E, Wise JH, et al. Rapid formation of massive black holes in close proximity to embryonic protogalaxies. *Nat Astron*. 2017 Mar;1: 0075.
- [80] Habouzit M, Volonteri M, Latif M, et al. On the number density of ‘direct collapse’ black hole seeds. *MNRAS*. 2016 Nov;463: 529–540.
- [81] Agarwal B, Cullen F, Khochfar S, et al. Optimal neighbourhood to nurture giants: a fundamental link between star forming galaxies and direct collapse black holes. *arXiv e-prints*. 2018 Aug. arXiv:1808.09981.
- [82] Johnson JL, Dijkstra M. Enhanced direct collapse due to Lyman α feedback. *A&A*. 2017 May;601: A138.
- [83] Spaans M, Silk J. Pregalactic black hole formation with an atomic hydrogen equation of state. *ApJ*. 2006 Dec;652: 902–906.
- [84] Ge Q, Wise JH. On the effect of Lyman α trapping during the initial collapse of massive black hole seeds. *MNRAS*. 2017 Dec;472: 2773–2786.
- [85] Grete P, Latif MA, Schleicher DRG, et al. Intermittent fragmentation and statistical variations during gas collapse in magnetised atomic cooling haloes. *arXiv e-prints*. 2019 Feb. arXiv:1903.00017.

- [86] Chon S, Hosokawa T, Yoshida N. Radiation hydrodynamics simulations of the formation of direct-collapse supermassive stellar systems. *MNRAS*. 2018 Apr;475: 4104–4121.
- [87] Schauer ATP, Regan J, Glover SCO, et al. The formation of direct collapse black holes under the influence of streaming velocities. *MNRAS*. 2017 Nov;471: 4878–4884.
- [88] Inayoshi K, Li M, Haiman Z. Massive black hole and population III galaxy formation in overmassive dark-matter haloes with violent merger histories. *MNRAS*. 2018 Sep;479: 4017–4027.
- [89] Greif TH, Bromm V, Clark PC, et al. Formation and evolution of primordial protostellar systems. *MNRAS*. 2012 Jul;424: 399–415.
- [90] Stacy A, Greif TH, Bromm V. The first stars: mass growth under protostellar feedback. *MNRAS*. 2012 May;422: 290–309.
- [91] Fialkov A, Barkana R, Tselikhovich D, et al. Impact of the relative motion between the dark matter and baryons on the first stars: semi-analytical modelling. *MNRAS*. 2012 Aug;424: 1335–1345.
- [92] Schauer ATP, Glover SCO, Klessen RS, et al. The influence of streaming velocities on the formation of the first stars. *MNRAS*. 2019 Apr;484: 3510–3521.
- [93] Hirano S, Hosokawa T, Yoshida N, et al. Supersonic gas streams enhance the formation of massive black holes in the early universe. *Science*. 2017 Sep;357: 1375–1378.
- [94] Becerra F, Marinacci F, Bromm V, et al. Assembly of supermassive black hole seeds. *MNRAS*. 2018 Nov;480: 5029–5045.
- [95] Luo Y, Ardaneh K, Shlosman I, et al. Direct collapse to supermassive black hole seeds with radiative transfer: isolated halos. *MNRAS*. 2018 May;476: 3523–3539.
- [96] Ardaneh K, Luo Y, Shlosman I, et al. Direct collapse to supermassive black hole seeds with radiation transfer: cosmological haloes. *MNRAS*. 2018 Sep;479: 2277–2293.
- [97] Wise JH, Regan JA, O’Shea BW, et al. Formation of massive black holes in rapidly growing pre-galactic gas clouds. *Nature*. 2019 Jan;566: 85–88.
- [98] Mayer L, Bonoli S. The route to massive black hole formation via merger-driven direct collapse: a review. *Rep Prog Phys*. 2019 Jan;82:016901.
- [99] Alexander T, Natarajan P. Rapid growth of seed black holes in the early universe by supra- exponential accretion. *Science*. 2014 Sep;345: 1330–1333.
- [100] Boekholt TCN, Schleicher DRG, Fellhauer M, et al. Formation of massive seed black holes via collisions and accretion. *MNRAS*. 2018 May;476: 366–380.
- [101] Katz H, Sijacki D, Haehnelt MG. Seeding high-redshift QSOs by collisional runaway in primordial star clusters. *MNRAS*. 2015 Aug;451: 2352–2369.
- [102] Gieles M, Charbonnel C, Krause MGH, et al. Concurrent formation of supermassive stars and globular clusters: implications for early self-enrichment. *MNRAS*. 2018 Aug;478: 2461–2479.
- [103] Li H, Gnedin OY, Gnedin NY, et al. Star cluster formation in cosmological simulations. I. Properties of young clusters. *ApJ*. 2017 Jan;834: 69.
- [104] Boylan-Kolchin M. The little engines that could? Globular clusters contribute significantly to reionization-era star formation. *MNRAS*. 2018 Sep;479: 332–340.
- [105] Li H, Gnedin OY. Star cluster formation in cosmological simulations. III. Dynamical and chemical evolution. *arXiv e-prints*. 2018 Oct. arXiv:1810.11036.
- [106] Johnson KE, Kobulnicky HA. The spectral energy distributions of infant super-star clusters in henize 2–10 from 7 Millimeters to 6 centimeters. *ApJ*. 2003 Nov;597: 923–928.
- [107] Clark JS, Negueruela I, Crowther PA, et al. On the massive stellar population of the super star cluster Westerlund 1. *A&A*. 2005 May;434: 949–969.
- [108] Vanzella E, Calura F, Meneghetti M, et al. Paving the way for the JWST: witnessing globular cluster formation at $z > 3$. *MNRAS*. 2017 Jun;467: 4304–4321.
- [109] Murray N, Quataert E, Thompson TA. The disruption of giant molecular clouds by radiation pressure & the efficiency of star formation in galaxies. *ApJ*. 2010 Jan;709: 191–209.
- [110] Skinner MA, Ostriker EC. Numerical simulations of turbulent molecular clouds regulated by reprocessed radiation feedback from nascent super star clusters. *ApJ*. 2015 Aug;809: 187.
- [111] Tsang BTH, Milosavljević M. Radiation pressure in super star cluster formation. *MNRAS*. 2018 Aug;478: 4142–4161.
- [112] Valiante R, Schneider R, Graziani L, et al. Chasing the observational signatures of seed black holes at $z > 7$: candidate statistics. *MNRAS*. 2018 Mar;474: 3825–3834.
- [113] Urry CM, Padovani P. Unified schemes for radio-loud active galactic nuclei. *PASP*. 1995 Sep;107: 803.
- [114] Steinborn LK, Hirschmann M, Dolag K, et al. Cosmological simulations of black hole growth II: How (in)significant are merger events for fuelling nuclear activity? *MNRAS*. 2018 Nov;481: 341–360.
- [115] Weinberger R, Springel V, Pakmor R, et al. Supermassive black holes and their feedback effects in the IllustrisTNG simulation. *MNRAS*. 2018 Sep;479: 4056–4072.
- [116] Wise JH, Abel T, Turk MJ, et al. The birth of a galaxy - II. The role of radiation pressure. *MNRAS*. 2012 Nov;427: 311–326.
- [117] Dijkstra M, Loeb A. Ly α -driven outflows around star-forming galaxies. *MNRAS*. 2008 Nov;391: 457–466.
- [118] Smith A, Bromm V, Loeb A. Lyman α radiation hydrodynamics of galactic winds before cosmic reionization. *MNRAS*. 2017 Jan;464: 2963–2978.
- [119] Kimm T, Haehnelt M, Blaizot J, et al. Impact of Lyman alpha pressure on metal-poor dwarf galaxies. *MNRAS*. 2018 Apr;475: 4617–4635.
- [120] Smith A, Ma X, Bromm V, et al. The physics of Lyman α escape from high-redshift galaxies. *MNRAS*. 2019 Mar;484: 39–59.
- [121] Smith A, Tsang BTH, Bromm V, et al. Discrete diffusion Lyman α radiative transfer. *MNRAS*. 2018 Sep;479: 2065–2078.
- [122] Smith A, Becerra F, Bromm V, et al. Radiative effects during the assembly of direct collapse black holes. *MNRAS*. 2017 Nov;472: 205–216.

- [123] Kormendy J, Ho LC. Coevolution (or not) of supermassive black holes and host galaxies. *ARA&A*. **2013 Aug**;51: 511–653.
- [124] Silk J, Rees MJ. Quasars and galaxy formation. *A&A*. **1998 Mar**;331: L1–L4.
- [125] Ferrarese L, Merritt D. A fundamental relation between supermassive black holes and their host galaxies. *ApJ*. **2000 Aug**;539: L9–L12.
- [126] Wehner EH, Harris WE. From supermassive black holes to dwarf elliptical nuclei: a mass continuum. *ApJ*. **2006 Jun**;644: L17–L20.
- [127] Graham AW, Spitler LR. Quantifying the coexistence of massive black holes and dense nuclear star clusters. *MNRAS*. **2009 Aug**;397: 2148–2162.
- [128] Nayakshin S, Wilkinson MI, King A. Competitive feedback in galaxy formation. *MNRAS*. **2009 Sep**;398: L54–L57.
- [129] Graham AW, Onken CA, Athanassoula E, et al. An expanded M_{bh} - σ diagram, and a new calibration of active galactic nuclei masses. *MNRAS*. **2011 Apr**;412: 2211–2228.
- [130] Shimasaku K, Izumi T. Black vs. dark: rapid growth of supermassive black holes in dark matter haloes at $z \sim 6$. arXiv e-prints. 2019 Feb. arXiv:1902.04165.
- [131] Haardt F, Madau P. Radiative transfer in a clumpy universe. IV. New synthesis models of the cosmic UV/X-ray background. *ApJ*. **2012 Feb**;746: 125.
- [132] Madau P, Haardt F. Cosmic reionization after Planck: could quasars do it all? *ApJ*. **2015 Nov**;813: L8.
- [133] Fialkov A, Barkana R, Cohen A. Reconstructing the nature of the first cosmic sources from the anisotropic 21-cm signal. *Phys Rev Lett*. **2015 Mar**;114:101303.
- [134] Planck Collaboration, Aghanim N, Akrami Y, et al. Planck 2018 results. VI. Cosmological parameters. arXiv e-prints. 2018 Jul. arXiv:1807.06209.
- [135] Stark DP, Ellis RS, Ouchi M. Keck spectroscopy of faint $3 > z > 7$ Lyman break galaxies: a high fraction of line emitters at redshift six. *ApJ*. **2011 Feb**;728: L2.
- [136] Schenker MA, Ellis RS, Konidaris NP, et al. Line-emitting galaxies beyond a redshift of 7: an improved method for estimating the evolving neutrality of the intergalactic medium. *ApJ*. **2014 Nov**;795: 20.
- [137] Mason CA, Fontana A, Treu T, et al. Inferences on the timeline of reionization at $z \sim 8$ from the KMOS lens-amplified spectroscopic survey. arXiv e-prints. 2019 Jan. arXiv:1901.11045.
- [138] Simcoe RA, Sullivan PW, Cooksey KL, et al. Extremely metal-poor gas at a redshift of 7. *Nature*. **2012 Dec**;492: 79–82.
- [139] Davies FB, Hennawi JF, Bañados E, et al. Quantitative constraints on the reionization history from the IGM damping wing signature in two quasars at $z > 7$. *ApJ*. **2018 Sep**;864: 142.
- [140] Oñorbe J, Hennawi JF, Lukić Z. Self-consistent modeling of reionization in cosmological hydrodynamical simulations. *ApJ*. **2017 Mar**;837: 106.
- [141] Kulkarni G, Keating LC, Haehnelt MG, et al. Large Lyman-alpha opacity fluctuations and low CMB tau in models of late reionization with large islands of neutral hydrogen extending to $z < 5.5$. arXiv e-prints. 2018 Sep. arXiv:1809.06374.
- [142] McGreer ID, Fan X, Jiang L, et al. The faint end of the $z = 5$ quasar luminosity function from the CFHTLS. *AJ*. **2018 Mar**;155: 131.
- [143] Puchwein E, Haardt F, Haehnelt MG, et al. Consistent modelling of the meta-galactic UV background and the thermal/ionization history of the intergalactic medium. *MNRAS*. **2019 May**;485: 47–68.
- [144] Eilers AC, Davies FB, Hennawi JF, et al. Implications of $z \sim 6$ quasar proximity zones for the epoch of reionization and quasar lifetimes. *ApJ*. **2017 May**;840: 24.
- [145] Eilers AC, Hennawi JF, Davies FB. First spectroscopic study of a young quasar. *ApJ*. **2018 Nov**;867: 30.
- [146] Natarajan P, Pacucci F, Ferrara A, et al. Unveiling the first black holes with JWST: multi-wavelength spectral predictions. *ApJ*. **2017 Apr**;838: 117.
- [147] Yue B, Ferrara A, Salvaterra R, et al. Infrared background signatures of the first black holes. *MNRAS*. **2013 Aug**;433: 1556–1566.
- [148] Pacucci F, Ferrara A, Volonteri M. Shining in the dark: the spectral evolution of the first black holes. *MNRAS*. **2015 Dec**;454: 3771–3777.
- [149] Kashlinsky A, Arendt RG, Atrio-Barandela F, et al. Looking at cosmic near-infrared background radiation anisotropies. *Rev Mod Phys*. **2018 Apr**;90:025006.
- [150] Helgason K, Ricotti M, Kashlinsky A, et al. On the physical requirements for a pre-reionization origin of the unresolved near-infrared background. *MNRAS*. **2016 Jan**;455: 282–294.
- [151] Dominik M, Belczynski K, Fryer C, et al. Double compact objects. II. Cosmological merger rates. *ApJ*. **2013 Dec**;779: 72.
- [152] Kinugawa T, Miyamoto A, Kanda N, et al. The detection rate of inspiral and quasi-normal modes of population III binary black holes which can confirm or refute the general relativity in the strong gravity region. *MNRAS*. **2016 Feb**;456: 1093–1114.
- [153] Inayoshi K, Hirai R, Kinugawa T, et al. Formation pathway of population III coalescing binary black holes through stable mass transfer. *MNRAS*. **2017 Jul**;468: 5020–5032.
- [154] Pacucci F, Loeb A, Salvadori S. Gravitational wave sources from pop III stars are preferentially located within the cores of their host galaxies. *MNRAS*. **2017 Oct**;471: L72–L76.
- [155] Belczynski K, Buonanno A, Cantiello M, et al. The formation and gravitational-wave detection of massive stellar black hole binaries. *ApJ*. **2014 Jul**;789: 120.
- [156] Volonteri M, Bellovary J. Black holes in the early Universe. *Rep Prog Phys*. **2012 Dec**;75:124901.
- [157] Ricarte A, Natarajan P. The observational signatures of supermassive black hole seeds. *MNRAS*. **2018 Dec**;481: 3278–3292.
- [158] Toma K, Yoon SC, Bromm V. Gamma-ray bursts and population III stars. *Space Sci Rev*. **2016 Dec**;202: 159–180.